Nikil PS

AI/ML Engineer | Cambridge, UK | nikilemuk@gmail.com | 07871075988 | LinkedIn | Github

Summary

AI Engineer with 2+ years of experience building production software across LLM applications, backend platforms, and embedded systems. Strong in Python, PyTorch, SQL, Docker, Kubernetes, and AI workflow orchestration, with hands-on experience building RAG pipelines, AI agents, internal automation platforms, and production services. MSc in Artificial Intelligence with practical experience in computer vision, inference workflows, and translating ML prototypes into deployable systems.

Skills and Certificates

Languages: Python, C, C++, JavaScript, SQL
Frameworks/Libraries: LangChain, FastAPI, PyTorch, Hugging Face, OpenCV, Scikit-learn, Pandas, LangGraph, A2A, MCP

Domains: Computer Vision, NLP, Embedded Systems, OCR
Tools: Git, Docker,Langsmith, Kubernetes, Linux, AWS

Certificates: NLP Specialisation, CAN and CAN FD, Embedded C,Generative AI Specialisation

Experience

Software Engineer, Multimatic Electronic System – Cambridge,GB — 2023 - Present

- Engineered a multimodal Python-based RAG and OCR pipeline for legacy document search, enabling structure-aware retrieval and higher-accuracy access to internal engineering knowledge.
- Developed internal AI agents for contextual search and automated requirement generation using company knowledge sources, reducing manual effort and improving process efficiency by 40%.
- Deployed an LLM agent on an edge device with access to internal tools via an MCP server, enabling secure communication and task completion in constrained environments.
- Designed and deployed a containerised Python key-generation service using Docker, Kubernetes, and CI/CD workflows, improving security compliance across 1,200+ vehicles.
- Built a full-stack internal automation platform with Python, Node.js, PostgreSQL, Svelte, and Jira API integrations to improve commit review workflows and traceability, delivering a 67% efficiency gain.
- Reworked and validated cryptographic RNG logic in C against NIST SP 800-22 standards to strengthen security assurance for 1200+ ECUs.

Education

University of Edinburgh, MSc Artificial Intelligence — Nov 2022

- MSc Dissertation: 83%

University of Manchester, BS(Hons) Mechatronics — June 2021

- First Class: 77%

Projects

Inference Engine | Pytorch MoE Inference System — GitHub Repo

- Built a PyTorch-based inference engine for Mixture-of-Experts models to reduce GPU memory pressure by loading only hot expert weights into VRAM during inference to optimise VRAM usage.
- Developed model inspection tooling that analyses GGUF tensor distributions and benchmarking workflows to compare CPU and GPU inference behaviour against baseline performance.

RAG Tool Selection Middleware | Agent Infrastructure — GitHub Repo

- Built a retrieval-based tool-selection layer for agentic workflows to scale tool usage across large tool libraries while reducing model context overhead and create a harness to measure performance of middleware.
- Indexed tool definitions through AST based tool extraction in a vector database and retrieved only the most relevant tools per turn to improve focus and support more scalable multi-step agent execution.

MSc Dissertation: Automated Recycling Sorting — University of Edinburgh

- Developed a real-time waste classification system using VarifocalNet on the MMDetection API, achieving high-speed inference and 5-class object detection.
- Improved model performance by increasing the mAP score by 244%, reducing training annotation by 70% and improving generalisability across object classes.